



10.3.6 A complete algorithm ¶

10.3.6 Putting it all together, Part 1

fit width



10.3.6 Putting it all together, Part 2

fit width



The last unit shows how one iteration of the QR algorithm can be performed on a tridiagonal matrix by implicitly shifting and then "chasing the bulge." All that is left to complete the algorithm is to note that

- The shift μ_k can be chosen to equal $\alpha_{m-1,m-1}$ (the last element on the diagonal, which tends to converge to the eigenvalue smallest in magnitude). In practice, choosing the shift to be an eigenvalue of the bottom-right 2×2 matrix works better. This is known as the **Wilkinson shift**.
- If $A = QTQ^T$ reduced A to the tridiagonal matrix T before the QR algorithm commenced, then the Givens' rotations encountered as part of the implicitly shifted QR algorithm can be applied from the right to the appropriate columns of Q so that upon completion Q is left overwritten with the eigenvectors of A . Let's analyze this:
 - Reducing the matrix to tridiagonal form requires $O(m^3)$ computations.
 - Forming Q from the Householder vectors requires $O(m^3)$ computations.
 - Applying Givens' rotation to a pairs of columns of Q requires $O(m)$ computation per Givens' rotation. For each Francis implicit QR step $O(n)$ Givens' rotations are computed, making the application of Givens' rotations to Q of cost $O(m^2)$ per iteration of the implicitly shifted QR algorithm. Typically a few (2-3) iterations are needed per eigenvalue that is uncovered (when deflation is incorporated), meaning that $O(m)$ iterations are needed. Thus, a QR algorithm with a tridiagonal matrix that accumulates eigenvectors requires $O(m^3)$ computation.

Thus, the total cost of computing the eigenvalues and eigenvectors is $O(m^3)$.

- If an element on the subdiagonal becomes zero (or very small), and hence the corresponding element of the superdiagonal, then

$$T = \left(\begin{array}{c|c} T_{00} & 0 \\ \hline 0 & T_{11} \end{array} \right) \quad \begin{array}{cccc|cccc} \times & \times & & & & & & \\ \times & \times & \times & & & & & \\ & \times & \times & \times & & & & \\ & & \times & \times & 0 & & & \\ & & & \times & \times & 0 & & \\ 0 & & & & & \times & \times & \\ & & & & & \times & \times & \times \\ & & & & & & \times & \times \end{array}$$

then

- The computation can continue separately with T_{00} and T_{11} .
- One can pick the shift from the bottom-right of T_{00} as one continues finding the eigenvalues of T_{00} , thus accelerating that part of the computation.
- One can pick the shift from the bottom-right of T_{11} as one continues finding the eigenvalues of T_{11} , thus accelerating that part of the computation.
- One must continue to accumulate the eigenvectors by applying the rotations to the appropriate columns of Q .

- Because of the connection between the QR algorithm and the Inverse Power Method, subdiagonal entries near the bottom-right of T are more likely to converge to a zero, so most deflation will happen there.
- A question becomes when an element on the subdiagonal, $\tau_{i+1,i}$ can be considered to be zero. The answer is when $|\tau_{i+1,i}|$ is small relative to $|\tau_i|$ and $|\tau_{i+1,i+1}|$. A typical condition that is used is

$$|\tau_{i+1,i}| \leq \epsilon_{\text{mach}} \sqrt{|\tau_{i,i}| + |\tau_{i+1,i+1}|}.$$

For details, see some of our papers mentioned in the enrichments.